



TobiasBelch

PRINCIPAL ENGINEER

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about

Principal Engineer with 17 years delivering web platforms in healthcare, e-health commerce, and sustainability. Defines technical strategy and architecture across team boundaries — owns stakeholder communication in high-visibility situations and mentors engineers to raise collective quality and delivery speed. Combines architecture ownership with hands-on TypeScript, React, and Next.js execution across frontend and backend layers. Drives Design System initiatives and AI agent integration at organisational scale.

skills

Architecture & Strategy

Frontend Architecture
BFF & API Design
Multi-Tenant Platforms
Event-Driven Systems
ADR & Design Reviews
Technical Strategy

Languages & Frameworks

TypeScript / JavaScript
React.js / Next.js
Node.js / NestJS
GraphQL / REST APIs
Material UI / Tailwind / SCSS

Platform & Quality

Jest / Cypress / Playwright
Accessibility Compliance
Docker / Git
D3 / Data Visualisation
AMQP / Apache Flink / NiFi

AI & Data

LLM Integration
Prompt Engineering
AI Agent Orchestration

Leadership

Technical Leadership
Organisational Influence
Mentoring & Career Growth
Stakeholder Management
Cross-Functional Alignment

languages

German (Native) · English (CEFR C-level)

work experience

Senior Software Engineer – DocMorris

Berlin, Germany | 08/2023 – Present

Leading digital healthcare and online pharmacy platform in the German market.

- Defined the architecture strategy for a white-label monolith-to-multi-tenant migration. Onboarded 3 brands across 11M active users with ~90% code reuse; second tenant launched in 2 months vs. an estimated 2 years without the white-label architecture — a ~92% time reduction. Refactoring improved testability and maintainability over the original monolith.
- Mentored 5 of 15 engineers through a structured program of 1:1s, pair programming, and weekly education sessions. One bootcamp-graduate engineer promoted to mid-level within 2 years and to senior within 4 — an estimated 2-year acceleration vs. industry average.
- Established LLM agent patterns for product creation workflows — piloted within own team, then packaged and handed to 3 additional teams. Improved response predictability and achieved an estimated 30x token-cost reduction on knowledge-retrieval tasks, with schema-validated outputs approaching deterministic reliability.
- Designed and led implementation of a cross-platform Design System (~30–40 components), structured from day one to enable AI-assisted UI generation. Projected to halve design time, eliminate handover overhead, and reduce implementation effort by an estimated 80% through AI-powered component assembly.
- Drove frontend and BFF architecture evolution, shaping technical direction across the stack.
- Trusted for stakeholder communication in critical and high-visibility situations.
- Improved test coverage and accessibility compliance across the frontend codebase.

Head of Engineering – Made2Flow

Berlin, Germany | 12/2022 – 07/2023

Sustainability platform for fashion supply-chain decision support.

- Co-designed a next-generation domain data model with the CPO, replacing an MVP-stage model that blocked ~30 business cases. Delivered the redesign in 6 months alongside other responsibilities, directly unblocking several client deals that were pending the transformation.
- Built and led a team of 4 engineers, hiring 2 within 2 months — including one of the strongest engineers I have worked with, who remains a key contributor. Established domain documentation and refinement processes that closed a gap between business goals and implementation, eliminating recurring friction.
- Owned technical partner communication and integration planning.

Technical Lead – eHealth-Tec GmbH

Berlin, Germany | 04/2020 – 11/2022

Digital health products for patients, doctors, and pharmacists.

- Replaced ad-hoc architecture decisions — where proposals were lost and arbitrary shifts caused chaos — with a structured ADR and design review process (~15–20 ADRs). Accelerated decision-making, improved codebase stability and maintainability, and reduced onboarding time for new engineers. Process still active 5 years on as the organisational standard.
- Architected BFF layers insulating the frontend (~11M users) from restructuring across ~20 backend services. Eliminated ripple-effect changes on the frontend whenever backend services were migrated or replaced, and provided the architectural seam for white-label tenant differentiation.
- Built and owned the frontend of a white-label online pharmacy platform serving 2 clients in production (TypeScript, React, Next.js).
- Led stakeholder communication and cross-functional alignment for a team of 4 in a Scrum-lite setup.
- Mentored and onboarded engineers; grew team capability in TypeScript, architecture, and testing.
- Delivered a gematik-regulated browser-based e-prescription tool for physicians.

work experience

Head of Engineering – Medtower UG

Berlin, Germany | 09/2018 – 04/2020

Software and hardware solutions for hospital operations and visibility.

- Architected a real-time distributed communication system using AMQP 1.0 / Apache Qpid where IoT hardware, servers, and web browsers operated as peers via pub/sub — eliminating point-to-point integration complexity and simplifying distributed system reasoning, monitoring, and debugging. The protocol's finance-sector provenance ensured the uptime and reliability required in a hospital environment.
- Built 3D indoor multi-floor maps with live device location for hospital equipment tracking (beds, wheelchairs, devices), including indoor navigation and reporting views. Entire system operated in an offline hospital intranet — no internet access — requiring fully self-contained tile serving and real-time updates.
- Managed an external Apache NiFi consultant for client-specific ETL pipelines. Within 2 weeks of engagement, the output enabled the team to build business cases.
- Owned architecture and development of a real-time IoT platform — microservices, SPA, and live device tracking (Node.js, React, TypeScript).
- Led a team of 2 engineers; established requirements engineering, refinement process, and team knowledge base.

earlier experience

Software Developer – remind me GmbH

02/2018 – 08/2018

Authored the technical concept for the white-label product transformation — adopted and shipped after tenure. Implemented feature-toggle architecture with lazy loading for flexible multi-context deployment. Mentored junior and mid-level developers in a Kanban team of 5.

Software Developer – Cedavis Technology GmbH

08/2009 – 01/2018

Evolved sole ownership of frontend architecture over 9 years — introduced Angular 1, then migrated to React and TypeScript as the stack matured. Built a custom D3 chart component library from scratch. Engineered a PDF reporting module using Java Nashorn to render React components server-side — enabling a single shared codebase for live UI and PDF output. Implemented data simulation and forecasting features for healthcare capacity planning.

certifications

iSAQB Certified Professional for Software Architecture – Foundation

05/2022

oase

Domain Driven Design

12/2021

entwickler.de

education

Berlin University of Applied Sciences (BHT)

2007 – 2011

Bachelor of Science, Computer Science

